Financial metrics and Gate B tests

Every metric we calculate and every test a strategy must pass: the formula, and how to read it.

> **Part 1** is what the Lightyear dashboard shows. **Part 2** is SE Return Analytics, the monthly block, which is now on the dashboard and in the workbook. **Part 3** is Gate B v3: four hard blocks and three scorecard rows, with the deflated Sharpe shown outside the gate.
>
> There are no run results or examples here. The only numbers are method settings and pass marks, checked against the code.

Notation used in every formula

Symbol	Meaning
r, b	One day's return on the strategy and on the benchmark, the NIFTY 500 price index.
m	One calendar month's return, compounded from the daily returns. Used in Part 2.
Rf (5%)	The risk-free rate in every Sharpe and Sortino. One config setting. A yardstick, not income.
c	The card's cash rate: what idle cash earns in the backtest. Beta and the factor regression subtract it. Not the risk-free rate.
V_t	Portfolio value on day t.
mean, stdev	Average and sample standard deviation (divides by n − 1). **Pop stdev** divides by n; Part 2 uses it.
√252, √12	Annualise daily and monthly volatility. 252 is the assumed trading days a year.
years	Calendar days ÷ 365.25. In Part 2, months ÷ 12.

Contents

Part 1. On the Lightyear dashboard **2**
- 1.1 The five headline numbers 2
- 1.2 The charts 2
- 1.3 Returns and risk table 2
- 1.4 Costs and execution table 4
- 1.5 Statistical robustness table 4
- 1.6 Run facts table 5

Part 2. SE Return Analytics (monthly) **6**
- 2.1 Performance 6
- 2.2 Risk Parameters 6
- 2.3 Performance Vs Index 7
- 2.4 Capture Ratio 7
- 2.5 Extreme Capture Ratio 8
- 2.6 Tail Risk 8
- 2.7 Period tables 8

Part 3. The Gate B v3 tests **9**
- 3.1 What a strategy must pass 9
- 3.2 Test 1: implementable within mandate 9
- 3.3 Test 2: data integrity 9
- 3.4 Test 3: edge over the benchmark 10
- 3.5 Test 4: paired stationary bootstrap 10
- 3.6 Test 5: sub-period stability 10
- 3.7 Test 6: factor regression 11
- 3.8 Test 7: performance after publication 11
- 3.9 Outside the gate: deflated Sharpe 11

Part 1. On the Lightyear dashboard

1.1 The five headline numbers

These tiles appear on every run's Results tab, for the strategy and the benchmark over the same days. The comparators table repeats CAGR, volatility, Sharpe and worst fall for each alternative.

CAGR: the steady yearly growth rate from start to finish

$$\text{CAGR} = \left(\frac{V_{\text{end}}}{V_{\text{start}}}\right)^{\frac{1}{\text{years}}} - 1$$

Volatility: how much returns swing in a year. Lower is a smoother ride

$$\text{Vol} = \text{stdev}\left(r_{\text{daily}}\right) \times \sqrt{252}$$

Sharpe (risk-free rate 5%): return above the risk-free rate, per unit of risk. Higher is better

$$\text{Sharpe} = \frac{\text{CAGR} - 5\%}{\text{Vol}}$$

The 5% is the risk-free rate, a yardstick. It is not income the strategy earns: idle cash earns the card's cash rate. This headline Sharpe is the geometric form. Figures for a single calendar year, and every Gate B test, use the arithmetic form shown in 1.3.

Worst fall from peak (maximum drawdown): the biggest loss from a high to a later low

$$\text{Max drawdown} = \max_{t}\left(1 - \frac{V_t}{\max_{s \le t} V_s}\right)$$

100 became: what 100 grew into, after every daily return

$$\text{Final} = 100 \times \prod_{t}\left(1 + r_t\right)$$

1.2 The charts

Chart	Calculation	How to read it
Growth of 100	100 × running product of (1 + r)	Above the benchmark's line means ahead of it on that day.
Drawdown	V_t ÷ highest V so far − 1	0% is a new high. Deeper dips are bigger losses.
Rolling volatility	stdev of the last 63 daily returns × √252	How bumpy the last three months were.
Calendar-year return	Product of (1 + r) within the year − 1	First and last years may be part-years.
Monthly heat map	Product of (1 + r) within the month − 1	Green months gained, red months lost.

1.3 Returns and risk table

The Core performance table is calculated by Lightyear for every run. The tables below it are written by each run's script, so **rows tied to a strategy's rules appear only for that strategy**. Each row names its source.

Metric	Formula	How to read it
Edge over benchmark	$CAGR_{strategy} - CAGR_{benchmark}$	Points a year ahead of the index.
Sortino (risk-free rate 5%)	(CAGR − 5%) ÷ DD, where DD = √mean(min(r − Rf per day, 0)²) × √252	Geometric, like the headline Sharpe. Only falls below the risk-free rate count as risk.
Sharpe, arithmetic	mean(r − Rf per day) ÷ stdev(r − Rf per day) × √252	The form used for each calendar year's Sharpe and in Gate B. It is labelled arithmetic wherever it appears.
Sortino, arithmetic	mean(r − Rf per day) × 252 ÷ DD	The yearly form of Sortino.
Calmar	CAGR ÷ maximum drawdown	Return per unit of worst fall.
Sub-period CAGR; worst sub-period excess	CAGR over part of the window; the smallest $CAGR_{strategy} - CAGR_{benchmark}$ across the windows	Whether the result depends on one stretch.
Drawdown in latest correction	Each series' fall over the index's latest fall deeper than 10%	How it behaved in the most recent real test.
Drawdown duration	Longest run of trading days below a previous high	How long the longest wait to recover took.
95% CVaR (daily loss)	−mean of the daily returns at or below their 5th percentile	The average bad day, as a positive number.
Skew	Sample skew of daily returns	Negative means big falls are more common than big rises.
CAGR without the 10 best days	Remove the 10 best days, then compound; years run from the first return date	Whether a handful of days made the result.
Beta	Cov(r − c, b − c) ÷ Var(b − c)	1.0 moves in line with the index. Below 1 is less sensitive.
Up-market / down-market beta	Beta using only days the index rose / fell	Ideally high in rises and low in falls. The two do not average to the overall beta.
Beta with the index filter open / shut	Beta using only days the card's index filter was open / shut. The filter's state on a day is read at the previous session's close	Shown when the card has an index filter. A low beta when shut is the filter holding cash.
Tracking error	stdev(r − b) × √252	How far it strays from the index. Used inside the information ratio.
Information ratio	mean(r − b) × 252 ÷ tracking error	Return over the index per unit of straying. Higher is better.
Up / down capture (monthly)	mean monthly return ÷ the benchmark's, in months the benchmark rose / fell	Above 1 up and below 1 down is the good shape.
Rolling one-year periods behind	Share of 252-day windows with compounded return below the index's	How often an investor would have been trailing.
Deflated Sharpe	See Part 3	Probability of skill after allowing for versions tried. Outside the gate.

1.4 Costs and execution table

Metric	Formula	How to read it
Two-way turnover a year	Σ |weight after − weight before| over every trade, ÷ years	1.0 means trading the whole portfolio's value once a year.
CAGR at another cost	Same book, re-simulated at that round-trip cost	An appendix sweep. The headline is at the cost on the card.
Edge at a high cost	CAGR at that cost − $CAGR_{benchmark}$	What survives expensive trading.
Capacity per Rs 1 crore	For each trade: (its share of capital × Rs 1 crore) ÷ the stock's 20-day average traded value. The 95th percentile P is shown, and the capital at which it reaches the limit: 10% ÷ P	Information only. No fund size is assumed, and the limit is a placeholder.
Fills; resize fills	Count of trades; count that only resized	How busy the book is.
Names held, mean	mean(stocks held per rebalance)	Too few means the result rests on few companies.
Rebalances with few names	Count with fewer than 5 names, overall and while buying was open	Separates a filter holding cash from a universe running dry.
Signals fully in cash; with few eligible	Counts of signals	How often filters left nothing or little to buy.
Names bought at first fill	Count of positions opened on day one	How the book starts.
Largest weight; weights above 25%	max(single weight); count of rebalances above 25%	Concentration risk.
Different stocks held	Count of distinct names	Higher means more breadth.
Top-2 share of gains	Gains of two best names ÷ gains of all winners	High means a couple of stocks made the result.
Mean / max cash share	mean / max(cash ÷ NAV)	How invested the book really was.
Signals with buying shut	Signals with the market filter closed ÷ all signals	How often the market filter stopped new buys.
Median holding period	Median calendar days, entry to exit	How long a position typically lasts.
Names removed by a rule	mean(top-band names rejected per signal)	How hard a filter bites.
Rule exits	Count of sells forced by each rule, and their mean return since entry	Whether a rule mostly cuts losers or winners.
Buys on big up days	Count of buys after a close up more than 9.5%	Proxy for buying at a price band, where a real fill may fail.
Sells on leaving the index	Count, and mean return since entry	What index exits cost or earned.
Small-cap share	mean(NAV share in Smallcap 250 names)	How much of the result is a small-cap bet.

1.5 Statistical robustness table

These feed or support the Gate B tests in Part 3, where the methods are explained. Every Sharpe here is arithmetic.

Metric	Formula	How to read it
Bootstrap Sharpe 95% range	2.5th and 97.5th percentile of Sharpe across the draws	A range that includes low values means the Sharpe is uncertain.
Paired P, each comparator	Share of draws where strategy Sharpe − comparator Sharpe ≤ 0	Small means the edge is hard to explain by chance. Test 4.
Paired P at a fixed block	The same P with the mean block fixed at 5, 21 and 63 days	A check that the answer does not hang on the block length. Reported, not judged.
Factor alpha a year	$(1 + \alpha)^{252} - 1$	Return left after known factors. Test 6.
Alpha t-stat (HAC)	α ÷ HAC standard error of α	How sure we are alpha is not zero. Test 6.
Factor alpha over the risk-free rate	The same regression with returns above Rf instead of c	Shown beside the gate's figure. Reported, not judged.
Factor loadings	β_k in the factor regression	How much of each factor it behaves like.
Factor R-squared	Share of variance the factors explain	High means mostly known factors.
Alpha with a factor left out; on a variant	The same regression, one factor removed or on a variant book	Shows how much one factor or rule was accounting for.
CAGR after one rule change	Re-simulate with one rule changed	A real effect keeps its sign when a rule moves.
CAGR with cash at another rate	Re-simulate with idle cash earning that rate	An appendix sweep: how much the cash rate contributes.

1.6 Run facts table

These are counts and dates that show what the test covered. They are not performance metrics.

Row	What it counts
Checks passed	Look-ahead and accounting checks that passed, out of those run.
Signals; trading days; first and last dates	How many decisions and days the run covers, and where it starts and stops.
Universe source; members; members priced; never priced	Which index list was used, and how many members could actually be traded. Never-priced members are a remaining survivorship risk.
Price moves that coincide with an adjustment change	Whether a price rule is tripped by splits and bonuses rather than real shocks.
Versions tried	N for the deflated Sharpe: versions from the paper, earlier runs and studies, plus this run's sensitivities.

Part 2. SE Return Analytics (monthly)

A separate block, on monthly returns. It is on the Results tab and in the workbook sheet "SE Return Analytics". Every figure uses calendar-month returns m compounded from the daily returns, the population standard deviation, and three inputs: risk-free rate 5%, Omega threshold 1% a month, drawdown limit 0%. Do not compare a figure here with a daily figure of the same name in Part 1. Gate B uses none of them. The benchmark is the price index.

Python and Excel calculate each metric separately. Python calculates it in ros/metrics/se.py; Excel calculates it from a live formula in the workbook. The workbook's "Python vs Excel" sheet sets the two side by side and marks each OK or FAIL at 1e-9.

2.1 Performance

Metric	Formula	How to read it
Total Return	∏(1 + m) − 1	What 1 became over the whole window.
CAGR	$(1 + \text{Total Return})^{12 \div \text{months}} - 1$	Twelve months count as a year, whatever the calendar days.
Avg Month Ret	mean(m)	A typical month.
Positive Months	count of months with m > 0	
Negative Months	count of months with m ≤ 0	A flat month counts as negative.
Max Month Ret	max(m)	
Min Month Ret	min(m)	
Best Month Contri	Max Month Ret ÷ Total Return	How much of the whole gain one month is worth.
Worst Month Contri	Min Month Ret ÷ Total Return	

2.2 Risk Parameters

Metric	Formula	How to read it
Annualized Volatility	pop stdev(m) × √12	Lower is smoother. Population stdev, on months.
Positive Monthly Vol	pop stdev of the months with m ≥ 0	Bumpiness of the winning months. Not annualised.
Negative Monthly Vol	pop stdev of the months with m ≤ 0	Bumpiness of the losing months. Not annualised.
Max Drawdown	min over months of (W ÷ highest W so far − 1), where W = running ∏(1 + m)	A negative number. **The peak starts at the first month's own value, so a loss in the first month is not counted.**
Average Annual Max Drawdown	mean of the Max Calendar Year Drawdown rows	A typical year's worst fall. The same first-month rule applies inside each year, and part-years are included.

Metric	Formula	How to read it
Sharpe Ratio	(CAGR − Rf) ÷ Annualized Volatility	Return above the risk-free rate per unit of risk.
Treynor Ratio	(CAGR − Rf) ÷ Beta	Return above the risk-free rate per unit of market risk.
Sortino Ratio	(CAGR − Rf) ÷ (Negative Monthly Vol × √12)	**Differs from the textbook:** the divisor is the spread of the losing months, not a downside deviation below a target.
Calmar Ratio	CAGR ÷ |Max Drawdown|	Return per unit of worst fall.
Sterling Ratio	CAGR ÷ (|Average Annual Max Drawdown| − drawdown limit)	**Differs from the textbook:** the usual 10-point adjustment is replaced by the drawdown limit, which is 0.
Omega Ratio	Σ(m − T) over months with m > T ÷ |Σ(m − T) over months with m < T|, T the monthly threshold	Above 1 means gains over the threshold outweigh shortfalls.
Gain to Pain ratio	Σ of months with m ≥ 0 ÷ |Σ of months with m < 0|	Winning months against losing months.
Tail Ratio	|95th percentile of m| ÷ |5th percentile of m|	Above 1 means big wins are larger than big losses.

2.3 Performance Vs Index

Metric	Formula	How to read it
Beta	slope of m on the benchmark's monthly returns	1.0 moves in line with the index.
Alpha	CAGR − $CAGR_{benchmark}$	Points a year ahead of the index.
Jensens Alpha (Adj Beta)	CAGR − $CAGR_{benchmark}$ × Beta	**Differs from its label:** no risk-free rate and no adjusted beta are used.
Positive Monthly Alpha	count of months with m > b	Months ahead of the index.
Negative Monthly Alpha	count of months with m < b	Months behind it.
Tracking Error	pop stdev(m − b) × √12	How far it strays from the index.
Information Ratio	Alpha ÷ Tracking Error	The CAGR gap, not the mean monthly gap, over tracking error.
Hit Rate	Positive Monthly Alpha ÷ (Positive + Negative Monthly Alpha)	Share of months ahead; ties are left out.
Max Monthly Alpha	max(m − b)	
Min Monthly Alpha	min(m − b)	

2.4 Capture Ratio

Metric	Formula	How to read it
Ret +Ve	mean(m) in months with b ≥ 0	What it earned in the index's up months.
Ret -Ve	mean(m) in months with b < 0	What it lost in the index's down months.

Metric	Formula	How to read it
Upside Capture	Ret +Ve ÷ the benchmark's Ret +Ve	Above 1 rises more than the index.
Downside Capture	Ret −Ve ÷ the benchmark's Ret −Ve	Below 1 falls less than the index.
Capture Ratio	Upside Capture ÷ Downside Capture	Above 1 gains more than it gives back.

2.5 Extreme Capture Ratio

Metric	Formula	How to read it
Ex. Ret +Ve	mean(m) in months with b ≥ 5%	The index's strongest months.
Ex. Ret -Ve	mean(m) in months with b < −5%	The index's worst months.
Ex. Upside Capture	Ex. Ret +Ve ÷ the benchmark's	
Ex. Downside Capture	Ex. Ret −Ve ÷ the benchmark's	
Extreme Capture Ratio	Ex. Upside Capture ÷ Ex. Downside Capture	Whether it holds up when the market moves hardest.

2.6 Tail Risk

Metric	Formula	How to read it
Skewness	sample skew of m	Negative means big falls are more common than big rises.
Kurtosis	sample excess kurtosis of m	A bell curve scores 0. Positive means fat tails.
95% VaR	5th percentile of m	A negative return, not a positive loss.
99% VaR	1st percentile of m	A negative return.
95% CVaR	mean of the months with m ≤ 95% VaR	The average month once you are in that tail.
99% CVaR	mean of the months with m ≤ 99% VaR	

2.7 Period tables

Metric	Formula	How to read it
Quarterly return	∏(1 + m) over the calendar quarter − 1	A quarter with fewer than three months is a part-period.
Financial Year Performance	∏(1 + m) from April to March − 1	* marks the first year, ** a last part-year.
Calendar Year Performance	∏(1 + m) from January to December − 1	Same marks.
Max Calendar Year Drawdown	The Max Drawdown formula on that year's months alone	**A loss in the year's first month is not counted**, because the peak starts at that month's own value.

Part 3. The Gate B v3 tests

3.1 What a strategy must pass

Each test rules out one way a backtest can fool you. The pass marks are fixed in config before the run, so nobody can move the bar after seeing the answer. Every test reads the run's daily returns at the cost on its approved card. **Hard blocks** decide the status. **Scorecard** rows are shown green, amber or red and do not block.

#	Test	Method	Pass mark	Counts as
1	Implementable within the fund's mandate	Rule check on actual positions	No breach	Hard block
2	Data integrity	Checks, card rules, priced coverage	All pass; coverage ≥ 90%	Hard block
3	Edge over the benchmark	CAGR gap	> 0	Hard block
4	Advantage over benchmark is significant	Paired stationary bootstrap	P ≤ 0.1	Hard block
5	Sub-period stability	Sharpe gap in 4 windows	≥ 50% of windows ahead; none below -1.0	Scorecard
6	Alpha survives the factor regression	Regression t-test with HAC errors	α > 0 and t ≥ 2.0	Scorecard
7	Performance after publication	Edge after the paper's date	Edge > 0%	Scorecard

> **Status is PASS or FAIL.** It is PASS when no hard block failed and tests 3 and 4 both ran and passed. The scorecard informs the reader and never blocks. **"Not run"** means a test had nothing to run on, for example no mandate limits on file. It is not a pass, it is listed beside the status, and it does not hold the gate up.
>
> **Tests 3, 4 and 6 are the statistical tests.** Test 5 is a stability test, test 7 asks whether the edge outlived publication, and tests 1 and 2 are practical checks. The deflated Sharpe (3.9) is outside the gate, so the gate does not test for selection luck.

Lightyear reads each row as PASS, FAIL, green, amber, red or not run. **"Evidence supports" on the Gate B tab is written by Claude, not calculated**, and the decision stays PENDING until a named person records it.

3.2 Test 1: implementable within mandate

The question. Could the fund actually hold these positions?

Method. A rule check on the positions actually held, not the settings: the largest cash share, the largest single weight and the lowest weight, against the fund's limits. A card cannot change the limits; an override needs a named person and a time, and is shown. **Pass:** no breach. **If it fails:** each breach is listed.

Watch out: the limits file must hold all three limits. If any is blank the row shows not run, which is not a pass.

3.3 Test 2: data integrity

The question. Can the numbers be trusted before any statistic is read?

$$\text{Coverage} = \frac{\sum \text{priced members} \times \text{sessions}}{\sum \text{all members} \times \text{sessions}}$$

Method. Three things from the run's own files: every look-ahead and accounting check the run wrote passed; Lightyear's audit of the card's rules found none broken; and enough index members had a price. **Pass:** all checks pass, no rule broken, coverage ≥ 90%. **If it fails:** the result rests on a broken check, a rule the run did not follow, or too many members with no price.

Watch out: members never priced are a survivorship risk even above the mark. They are named in the evidence.

3.4 Test 3: edge over the benchmark

The question. Did it beat the index at all?

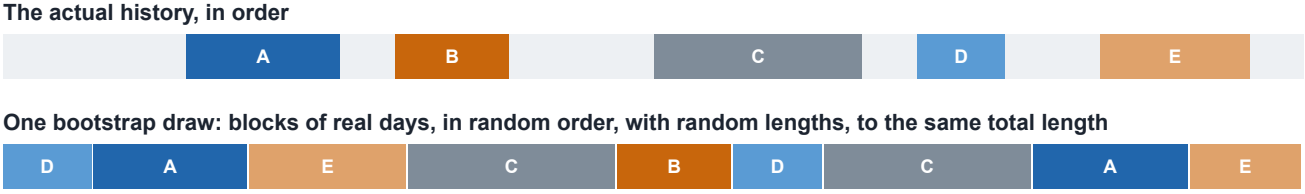
$$\text{Edge} = \text{CAGR}_{\text{strategy}} - \text{CAGR}_{\text{benchmark}}$$

Method. Both CAGRs over the same days, at the cost on the card. **Pass:** edge > 0. **If it fails:** the strategy did not beat the index it is measured against.

Watch out: on a card set to zero cost this is before any trading cost, so read the turnover and the cost sweep. The benchmark is a price index, which flatters the edge.

3.5 Test 4: paired stationary bootstrap

The question. We cannot re-run history. If we rebuild alternative histories from the real one, how often does the edge over the benchmark disappear? (Politis and Romano, 1994.)



The strategy and the benchmark are cut at the same days on each draw. That is what "paired" means.

- **Build:** pick a random start day and take a block of days, with random lengths, until the history is full length. Blocks keep calm and stormy spells together. The average block length is chosen from the data (Politis and White, 2004), as the largest of the lengths for the strategy, the benchmark and their difference.
- **Pair:** cut the strategy and the benchmark at the same days, which preserves their correlation.
- **Score:** calculate each arithmetic Sharpe, mean(r − Rf) ÷ stdev(r − Rf) × √252 with Rf at 5%, on that history, and take the gap D.
- **Repeat** 2,000 times with fixed seed 7, so the result can be reproduced.

$$D = \text{Sharpe}_{\text{strategy}} - \text{Sharpe}_{\text{benchmark}} \qquad P = \frac{\text{number of draws with} D \leq 0}{\text{number of draws}}$$

Pass: P ≤ 0.1. **If it fails:** the gap over the index could be noise. **Watch out:** P is a share of re-shuffled histories, not the chance the strategy has no edge. Values just either side of the mark are nearly the same evidence. It looks at Sharpe only, so a shallower drawdown does not help. P at fixed blocks of 5, 21 and 63 days is shown beside it and is not judged.

3.6 Test 5: sub-period stability

The question. Did it work in every stretch, or did one period make all the profit?

$$\text{Gap}_w = \text{Sharpe}_{\text{strategy},w} - \text{Sharpe}_{\text{benchmark},w}$$

Method. Split the run into 4 equal windows that do not overlap, and take the arithmetic Sharpe gap to the benchmark in each, with Rf at 5%. Windows under 60 days are dropped. If the card says the rules fit parameters to data, the first 5 years are skipped and the test is a walk-forward. **Pass:** green when at least 50% of windows

have a gap above 0 and no gap is below -1.0; amber while the worst gap is at least -1.5; otherwise red. **If it fails:** the edge depends on when you measured it.

Watch out: it is judged against the benchmark, so a weak market does not fail it. If the rules fit no parameters, these are sub-periods, not out-of-sample tests.

3.7 Test 6: factor regression

The question. Once you remove what known factors explain, is any return left? If not, you are paying for research to get exposure you can buy off the shelf.

$$r - c = \alpha + \sum_k \beta_k \left(f_k - c \right) + \varepsilon \qquad t = \frac{\alpha}{\mathrm{SE}_{\mathrm{HAC}}(\alpha)} \qquad \mathrm{lags} = \left\lfloor 4 \left(\frac{n}{100} \right)^{2/9} \right\rfloor$$

Method. Regress daily strategy returns on 7 factor index returns, all above the card's cash rate: NIFTY 500, NIFTY MIDCAP 150, NIFTY SMALLCAP 250, NIFTY500 MOMENTUM 50, NIFTY500 VALUE 50, NIFTY500 QUALITY 50, NIFTY500 LOW VOLATILITY 50. α is the unexplained return and β_k the loading on factor k. The t-stat is α divided by its uncertainty. HAC (Newey and West, 1987) is a safer standard error, since daily returns depend on each other and volatility clusters. The lag formula sets how many days of dependence it allows. **Pass:** green when α > 0 and t ≥ 2.0; amber from t ≥ 1.0; otherwise red. The mark is one-sided: a negative alpha cannot be green. **If it fails:** no reliable return is left after known factors.

Watch out: a factor left out of the regression can make alpha look larger, and the factor indices are backfilled and price-only. The test subtracts the card's cash rate, not the risk-free rate.

3.8 Test 7: performance after publication

The question. Did the edge survive once the idea was public?

$$\mathrm{Edge}_{\mathrm{after}} = \mathrm{CAGR}_{\mathrm{strategy}} - \mathrm{CAGR}_{\mathrm{benchmark}}, \text{ from the publication date}$$

Method. The same edge as test 3, from the paper's publication date onward. A date given only as a year is read as 31 December of that year. It needs at least 252 trading days after that date; with fewer the row is not run. **Pass:** green when the edge is above 0%; amber above -2%; otherwise red. **If it fails:** the result leaned on the years before anyone could have read the paper.

Watch out: a short stretch after publication is weak evidence either way.

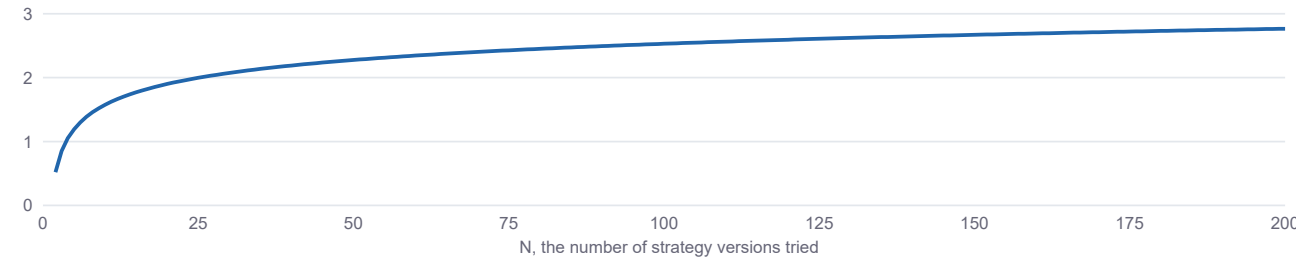
3.9 Outside the gate: deflated Sharpe

The question. Try enough versions of a strategy and one will look good by luck. Is this Sharpe higher than the best you would expect from that many versions with no skill, given how noisy and fat-tailed the returns are? (Bailey and López de Prado, 2014.) It is shown with each run and does not affect the status.

Step 1. How high is the luck bar?

$$\mathrm{SR}_0 = \sqrt{V} \times \left[\left(1 - \gamma \right) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

The luck bar SR_0 rises with N (shown with the default V = 1)



Step 2. How likely is it that the true Sharpe clears that bar?

$$\mathrm{P(skill)} = \Phi\left(\frac{\left(\mathrm{SR} - \mathrm{SR}_0\right)\sqrt{n-1}}{\sqrt{1 - S \cdot \mathrm{SR} + \frac{K-1}{4} \cdot \mathrm{SR}^2}} \right)$$

Input	Meaning
N	Strategy versions tried: the versions the paper reports, every distinct version run before, earlier studies on the trial ledger, plus this run's sensitivities. **This is a count of versions, not of tests.** A re-run that changes no trade is not a new version.
V	Variance of the versions' yearly Sharpes. The bar scales with √V, so the result is very sensitive to it. Two answers are shown: V = 1, the default, which sets a high bar, and V measured on this run's variants, which sets a very low one.
SR, SR_0	Observed Sharpe (arithmetic, over the risk-free rate of 5%) and the luck bar, both per day (÷ √252).
n	Number of daily returns. More history means more certainty.
S, K	Skew and kurtosis, where a bell curve has K = 3. Negative skew and fat tails make a given Sharpe less certain.
Φ, Φ^{-1}; γ, e	The bell-curve probability function and its inverse; the constants 0.5772 and 2.718.

Why it is outside the gate. With V = 1 it fails almost everything; with V measured on near-identical variants it passes almost everything. Until V is settled it is reported for the reader and does not decide the status. **Watch out:** versions are rarely independent, which overstates the penalty.

> **Not part of Gate B.** The monthly block in Part 2, and the capacity figure in 1.4, are information only. run_pipeline.py also has a promotion ladder that reuses some of these statistics at different bars. Lightyear does not use the ladder.

Sources: ros/metrics/core.py, ros/metrics/se.py, lightyear/charts.py, lightyear/gate_run.py, lightyear/workbook.py, lightyear/se_sheet.py, ros/governance/gate_v2.py, ros/validation/gate_evidence.py, research.py, portfolio.py, trials.py, config/fund.yaml, config/gate_thresholds.yaml, config/mandate.yaml. Bailey and López de Prado (2014), *Journal of Portfolio Management* 40(5). Politis and Romano (1994), *JASA* 89(428). Politis and White (2004), *Econometric Reviews* 23(1). Newey and West (1987), *Econometrica* 55(3).